When Is Voice Not Enough? A Deployment-Scale Analysis of Multimodal Necessity in Real-Time Speaker Attribution

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***Abstract*—Multimodal speaker attribution systems are typically evaluated with a single aggregate accuracy or coverage figure, which implicitly assumes that the benefit of fusing voice, face, and active-speaker cues is uniform across deployment conditions. We test this assumption using deployment logs from a real-time meeting-tracking system that attributes each speech segment through an explicit, priority-ordered audio-visual fusion cascade rather than a single learned model. Across 92 meeting sessions and 1,660 speech segments collected over two months, we find that reliance on voice biometrics varies from 5% to 40% of segments depending on expected group size, while visual and session evidence extends coverage to every accepted segment. The cascade sustains real-time throughput (89.6% of segments faster than real time). In a human-reviewed subset, its risk flag separates unflagged named decisions (81.8% exact agreement) from flagged ones (24.8%; Fisher's exact $p < 0.001$), effectively prioritizing review.**

***Index Terms*—biometrics, speaker recognition, feature extraction, feature matching and inference, active speaker detection, real-time computer vision, audio-visual fusion**

I. Introduction

Multimodal speaker attribution combining voice biometrics, face re-identification, and active speaker detection (ASD) is generally reported to outperform any single modality [1]–[3]. Almost all such evaluations, however, report one aggregate figure per system, computed over a benchmark whose composition (how many people are in frame, how long the deployment has been running, how well individual speakers are enrolled) is fixed by construction. A meeting-transcription system deployed in practice sees none of that stability. Group size varies from a one-on-one call to a full room, and the same system may run for months as its voice bank grows and its thresholds get tuned. Whether the reported benefit of fusion holds uniformly across these conditions, or is concentrated in some and nearly absent in others, is a question that a single aggregate figure cannot answer.

We study this question using deployment logs from a real-time, single-machine multimodal meeting tracker that attributes each speech segment through an explicit *priority cascade*, an ordered sequence of decision rules in which the first rule whose evidence requirement is met determines the final identity, rather than a single learned fusion model. At its core, the underlying problem is a multimodal *biometric feature matching and inference* task, in which voice and face embeddings are extracted independently by two different feature-extraction pipelines (ECAPA-TDNN acoustic embeddings and EdgeFace visual embeddings) and must be matched, cross-checked, and reconciled online against a per-session identity registry. We use the cascade's logged decisions, collected over 92 real, unscripted sessions and 1,660 speech segments spanning just over two months, as an instrument to measure how the necessity of each modality shifts across deployment conditions rather than to report one more aggregate accuracy number.

Our contributions are the following.

1) Evidence that reliance on voice biometrics is highly non-uniform across group sizes, ranging from 5% to 40% of segments, with a structural explanation tied to which cascade rule takes precedence at each group size, rather than a smooth trend.
2) A demonstration that an apparent decline in voice-led resolution over a multi-month deployment is substantially confounded by a shift in session composition, a methodological caution for interpreting raw temporal trends in deployment logs without controlling for this kind of shift.
3) A statistically tested human validation of the lightweight self-diagnostic risk monitor, together with a real-time performance characterization of the cascade under deployment conditions.
4) A reproducible evaluation protocol, with anonymized segment-level tables and scripts that regenerate the reported metrics.

II. Related Work

Speaker diarization and verification. pyannote.audio [4], [5] provides the diarization backbone used in our pipeline, segmenting and clustering speech into per-chunk speaker labels; diarization alone does not solve identity persistence across non-adjacent processing windows. ECAPA-TDNN [6], trained on VoxCeleb, is the standard architecture for the fixed-dimensional speaker embeddings we use both for enrolled-speaker matching and for a persistent within-session tracker for unenrolled speakers.

Active speaker detection. SyncNet-style lip-sync models [1] and TalkNet [2] established the audio-visual correlation approach to ASD; Light-ASD [3] is a lightweight CVPR 2023 model designed for real-time inference, which we use as the primary ASD backend. We use YOLOv8 [7] for face detection, ByteTrack [8] for short-term tracking, and EdgeFace [9] for lightweight face-recognition embeddings.

Audio-visual fusion for speaker identity. Most prior work on audio-visual speaker identification targets offline, dataset-level fusion, such as re-scoring diarization with ASD or jointly training audio-visual embeddings, and reports a single accuracy figure over a benchmark of fixed composition. Our setting differs in requiring an *online*, segment-by-segment decision under partial observability, where a face, a voice match, or both may be missing or unreliable for any given segment, which led us to a rule-based, explicitly ordered fusion strategy rather than a single learned fusion score.

Generalization across deployment conditions. Benchmarks such as AVA-ActiveSpeaker [10] and VoxConverse [11] fix group size and recording conditions by construction, so a system's reported accuracy is, by design, an average over whatever mixture of conditions the benchmark happens to contain. We are not aware of prior work that stratifies a deployed multimodal attribution system's behavior by group size or by deployment time to test whether a single aggregate figure is representative, which is the gap this paper addresses.

III. System Overview

The system runs two cooperating threads sharing an in-memory state object. A *video thread* performs face detection (YOLOv8), tracking (ByteTrack), face re-identification (EdgeFace), and per-frame ASD updates. An *audio thread* performs voice activity detection, diarization-based session hints (pyannote.audio), transcription (faster-whisper [12], Portuguese), and speaker verification (ECAPA-TDNN) on each detected speech segment, querying the video thread's state for currently visible faces and ASD results. When diarization indicates more than one concurrent speaker, the system applies SepFormer [13] speech separation before re-verifying each separated stream.

Algorithm 1 summarizes the fusion cascade that attributes each segment. Each rule has a different evidence requirement, and the cascade stops at the first rule whose precondition is satisfied. It prefers voice biometrics whenever the match is confident enough, since a voice match is invariant to camera framing and remains available for off-camera speakers, and falls back to progressively weaker, more circumstantial evidence otherwise. Two cross-cutting passes run afterward. A **gender-consistency check** rejects or reroutes visually derived decisions that contradict an unambiguous audio-derived gender signal, skipped when voice-biometric confidence is very high (≥ 0.90). A **turn-merging** pass merges consecutive same-diarization-label segments only if their voice-embedding similarity exceeds 0.40, since diarization occasionally mislabels two different speakers under the same local cluster.

The two accepted-match tiers in Algorithm 1 are separated by the confirmation threshold τ_c, which is configured independently of the acceptance threshold τ. The deployment ran at $\tau = 0.80$, $\tau_c = 0.70$ for 97% of segments, and at that operating point every accepted match also clears τ_c, so the two tiers collapse into one and VERIFIER-MOD is reached only in a small set of sessions configured with a looser $\tau = 0.55$. Because this split is an artifact of configuration rather than of evidence strength, we report all three verifier tiers together as a single voice-led decision family throughout §V.

Algorithm 1 Speaker attribution cascade (simplified)

$v \leftarrow$ accepted ECAPA match with confidence c; accepted only if $c \geq \tau$ and its margin over the runner-up is ≥ 0.07
$r \leftarrow$ best raw ECAPA candidate (name, confidence), accepted or not
$g \leftarrow$ audio-derived gender estimate
$a \leftarrow$ ASD query over [segment start $-$ 0.4s, end $+$ 0.4s]
if v is not NONE, $c \geq \tau_c$, **and** gender-consistent **then**
  return VERIFIER-HIGH (v.name)
else if v is not NONE **and** gender-consistent **then**
  return VERIFIER-MOD (v.name)
else if r.conf ≥ 0.65 **and** (ASD or gender corroborates) **then**
  return VERIFIER-WEAK (r.name)
else if a is not NONE **then**
  return ASD (face bound to a)
else if exactly 1 face visible, known identity, $\leq$1 expected speaker **then**
  return FACE-ONLY
else if exactly 1 face visible, r.conf ≥ 0.35, $\leq$1 expected speaker **then**
  return SINGLE-FACE (heuristic)
else if ASD best-guess is dominant ($\geq$1.5$\times$ runner-up) **then**
  return ASD-GUESS
else if diarization session hint bound to a known face **then**
  return SESSION-FACE
else if diarization session hint available **then**
  return SESSION-HINT (anonymous)
else
  return SESSION-TRACKER (persistent voice cluster)
end if
(Gender cross-check / reroute applied as a final pass)

ASD itself uses a two-backend design, with Light-ASD [3] as the primary backend and a model-free pixel-difference fallback for windows in which a neural score is not yet available. This design preserves continuous online operation. The fallback also rejects occlusion-related motion by requiring the mouth region to move at least 1.8$\times$ more than the upper face before the motion is treated as speech-related.

IV. Experimental Setup

The evaluation uses an internal development dataset collected during real, unscripted operation from 16 March to

22 May 2026. A session was included when it contained at least one accepted speech segment. This criterion retained 92 sessions and 1,660 segments; 14 empty sessions were excluded before analysis. The sessions were recorded in Portuguese with consumer webcam and microphone hardware under varying lighting, framing, overlap, and enrollment availability. Their configured group-size distribution was 19 one-speaker, 7 two-speaker, 4 three-speaker, and 62 four-speaker sessions. Processing used one machine with an Intel Core i5-11400H, 8 GB RAM, and an NVIDIA RTX 3060 Laptop GPU with 6 GB VRAM.

Audio is captured at 16 kHz. The first 8 s of a session form the initial processing window; subsequent windows contain 15 s and advance by 5 s, preserving 10 s of context while transcribing only the new region. pyannote.audio 3.1 receives the configured number of speakers as both its minimum and maximum, uses clustering threshold 0.60 and segmentation threshold 0.40, and retains turns of at least 0.5 s. The transcription stage uses beam size 5, speech VAD, a 500 ms minimum silence duration, and 300 ms speech padding. The model configured for each retained session was recorded in the database: 87 sessions used `large-v3-turbo`, 3 used `large-v3`, and 2 used `medium`. This distribution is preserved in the analysis rather than treating the deployment as a single-model benchmark.

TABLE I
PRINCIPAL OPERATING CONFIGURATION.

Stage	Configuration
Audio	16 kHz; 8 s initial; 15 s window; 5 s step
Diarization	pyannote 3.1; 0.60 clustering; 0.40 segmentation
Transcription	faster-whisper; beam 5; VAD enabled
Voice match	$\tau = 0.80$ (88 sess.); 0.55 (4); margin 0.07
Confirmation	$\tau_c = 0.70$ in all 92 sessions
ASD query	speech interval expanded by ± 0.4 s

For every accepted segment, SQLite stores the session identifier, segment index, final decision type, identity, raw voice confidence, ASD and face indicators, risk flag, speech duration, and transcription, verification, and total processing times. We group decisions into five families: voice-only, voice with corroboration, active-speaker, face or gender, and session context. This segment-level provenance makes each aggregate in the results traceable to the rule that produced it.

The five evidence families are mutually exclusive and exhaustive. VERIFIER-HIGH and VERIFIER-MOD form the voice-only family; VERIFIER-WEAK forms voice with corroboration; ASD and ASD-GUESS form active-speaker evidence; face and gender decisions form the visual identity family; and SESSION-FACE, SESSION-HINT, and SESSION-TRACKER form session context. This mapping prevents a segment from contributing to more than one percentage in the aggregate coverage analysis.

For segment i, with total processing time t_i in milliseconds and speech duration d_i in seconds, the real-time factor is

$$\mathrm{RTF}_i = \frac{t_i}{1000 d_i}. \tag{1}$$

An RTF below 1 means that processing completed faster than the speech segment itself. We report the median RTF and the proportion of segments with $\mathrm{RTF} < 1$, together with stage-level medians.

Identity agreement was evaluated on 330 manually reviewed segments from 24 sessions. When more than one review existed for a session, only the latest was retained. Exact agreement requires equality between the predicted and corrected identity after whitespace and case normalization. The named subset excludes anonymous diarization labels. Proportions are accompanied by 95% Wilson intervals [14]; the association between the risk flag and exact agreement is tested with a two-sided Fisher exact test at $\alpha = 0.05$.

DER and WER require time-aligned speaker turns and reference word transcripts, respectively [15]. Those annotations were not part of the deployment log, and the fusion cascade neither changes the Whisper words nor emits a reference-aligned diarization timeline. We therefore report metrics aligned with the evaluated task: decision provenance, exact named-identity agreement, and RTF, without presenting component metrics as end-to-end attribution scores.

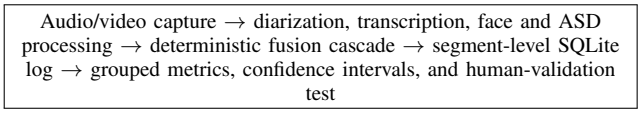


Fig. 1. Evaluation flow from multimodal capture to reproducible metrics.

The anonymous artifact at https://github.com/anonimo826/anonimo contains the system snapshot, parameter file, data dictionary, privacy-preserving derived tables, and an analysis script that regenerates the counts, Wilson intervals, RTF statistics, and Fisher test with one command. The released tables use sequential session codes and binary identity-agreement fields; they do not contain audio, video, names, transcripts, absolute dates, or biometric embeddings. SHA-256 hashes and a manifest allow researchers to verify that the analyzed artifacts remain unchanged.

V. RESULTS

A. Voice-led resolution is non-monotonic in group size

Across the whole deployment, voice evidence leads 8.6% of the 1,660 decisions: 4.0% are accepted voice matches and 4.6% are voice matches corroborated by ASD or gender. The remaining 91.4% demonstrate the coverage gained from ASD, face re-identification, gender consistency, and persistent session context. Table II stratifies the same segments by the configured group size. The voice-led rate is not a smooth function of group size, and each value has a structural explanation rooted in the cascade itself (Algorithm 1), not merely in how hard voice matching happens to be. Single-speaker sessions show a low voice-led rate (11.3%, 95% CI [5.8, 20.7]) because the cascade's face-identity rules are unlocked precisely when at most one speaker is expected, so voice matching is rarely the deciding rule even when it would have agreed. Two-speaker sessions show the highest voice-led rate observed (39.8%, 95%

CI [31.5, 48.8]), plausibly because a two-person voice bank is the easiest case to disambiguate and these sessions had the highest proportion of fully enrolled participants. Three-speaker sessions lean heavily on the gender-consistency layer (23.3%) and, as Table III shows below, also carry the highest self-flagged risk of any group size. Four-speaker sessions, which contribute 82% of all segments, drop to the lowest voice-led rate (5.0%, 95% CI [3.9, 6.3]) while ASD carries the largest share (36.8%). Together, these results show that the cascade adapts its evidence source to session composition and that the visual branches are most active in the dominant four-speaker setting.

TABLE II
DECISION SOURCE BY NUMBER OF EXPECTED SPEAKERS PER SESSION. COLUMNS REPORT THE THREE MAIN DECISION FAMILIES AND DO NOT SUM TO 100%; THE BALANCE IS CARRIED BY THE FACE-ONLY, SESSION-HINT AND SESSION-TRACKER RULES.

Speakers	Sess.	Segs.	Voice	ASD	Gender
1	19	71	11.3%	16.9%	1.4%
2	7	118	39.8%	35.6%	0.8%
3	4	103	19.4%	22.3%	23.3%
4	62	1368	5.0%	36.8%	6.9%

Table III cross-checks this with the self-diagnostic risk flag (§V-E). The risk rate is also non-monotonic and peaks at three speakers (68.9%), well above both its neighbors, matching the group size where the gender-safety layer had to work hardest in Table II. Three-speaker meetings therefore receive the highest review priority from the monitor. This behavior is consistent with the strong participation of the gender-consistency layer in the same group.

TABLE III
SELF-FLAGGED REVIEW-PRIORITY RATE BY EXPECTED SPEAKERS.

Speakers	Sess.	Segments	Flagged %
1	19	71	23.9%
2	7	118	5.9%
3	4	103	68.9%
4	62	1368	28.1%

B. Apparent temporal decline is a composition effect

A naive reading of the deployment log by calendar week shows what looks like a steep decline in voice-led resolution. Voice-led decisions accounted for 37–42% of segments in the first two weeks, falling to 0–2% in the final three weeks recorded. Read on its own, this trend invites a misleading story, namely that voice matching degraded as the deployment progressed.

Breaking down the same weeks by session composition rules this out. The first two weeks were composed entirely of one- and two-speaker sessions (100% in each week), the group sizes Table II identifies as having the highest voice-led rates. From the third week onward, the deployment shifted to sessions with three or four expected speakers, reaching 90% four-speaker segments in the third week and staying above 80% in every subsequent week with more than 100 segments. The decline in voice-led resolution over time is, to a substantial degree, the same non-monotonic group-size effect from §V-A appearing again under a different label, because the deployment's session mix changed, not because voice biometrics got worse at its job.

The self-flagged risk rate over the same weeks makes the point a second time, and in the opposite direction. It was zero in the first two weeks, rose to 39% and then 52% as multi-speaker sessions came to dominate, and fell back to 14% in the final week. Read as a time series alone, the rise and the fall would each invite a story about the system changing. Both are largely the composition shift again, with the final-week drop additionally consistent with threshold and gender-check refinements made late in the period, a genuine improvement running in parallel with, and partly obscured by, that shift. We take this as a methodological caution. A raw trend line over a real deployment log can be confounded by what the deployment happened to contain in each period, and stratifying by a plausible confounder, here group size, before drawing conclusions from time alone is necessary, not optional.

C. Real-time performance

The cascade processed 1,488 of 1,660 segments (89.6%) faster than real time, with median RTF 0.24. Median total processing time was 633.1 ms per segment; transcription accounted for 513.6 ms and voice verification for 29.7 ms. These measurements show that multimodal attribution remains compatible with live meeting operation on a single consumer laptop GPU.

D. ASD backend accuracy against the published literature

The ASD backend is used exactly as released, with no retraining or fine-tuning on our data, so the figures its authors report describe the same weights that run in our pipeline. The pretrained Light-ASD checkpoint we use is reported by its authors to reach 94.06% mAP on the AVA-ActiveSpeaker validation set and an average F1 of 81.1% on the Columbia ASD dataset [16], a small, five-panelist benchmark widely used for out-of-domain ASD evaluation [3]. Both are fixed-composition benchmarks of the kind this paper's stratified deployment analysis is meant to complement, and neither speaks to how often ASD is the rule that decides an identity, which is what Table II measures.

E. Statistical validation of the review-priority monitor

Independently of the cascade, the system computes a rule-based risk flag for every segment that received a named identity. Four factors raise it. A decision of a structurally weak type (VERIFIER-WEAK, FACE-ONLY, SINGLE-FACE) raises its review priority, as does the number of distinct identified speakers in the session exceeding the configured expected count. A name drawn from the enrolled voice bank whose raw voice confidence never reached the acceptance threshold τ receives high priority, as does a contradiction between the attributed name's gender and the audio-derived gender

estimate. The monitor is a deterministic function of decision metadata and therefore operates without labels. Across the deployment, 479 of 1,660 segments (28.9%) were flagged for priority review.

The human-reviewed subset confirms that this prioritization is informative. Among named decisions, 9 of 11 unflagged cases agreed with the human correction (81.8%, Wilson 95% CI [52.3, 94.9]), compared with 28 of 113 flagged cases (24.8%, 95% CI [17.7, 33.5]). Fisher's exact test rejects independence between the flag and exact agreement ($p = 2.94 \times 10^{-4}$). Thus, the monitor identifies a substantially different review-priority group and provides a practical way to focus human verification effort.

VI. Discussion

The results support a condition-aware view of multimodal attribution. The same cascade moves naturally between voice, ASD, face, and session context as group composition changes. This is particularly valuable in four-speaker sessions, which represent 82.4% of the evaluated segments: visual or session evidence participates in 95.0% of their decisions, preserving coverage when voice alone is not decisive.

The combination of traceable decision rules, a statistically validated review-priority monitor, and real-time execution offers three practical benefits. Each identity can be audited back to its evidence source, segments can be ranked for efficient human review, and the complete analysis can be regenerated from privacy-preserving tables. These properties make the system a reproducible foundation for future condition-aware threshold selection and larger annotated evaluations.

VII. Conclusion

In 92 real meeting sessions and 1,660 speech segments, the system combined complementary modalities while sustaining real-time operation for 89.6% of segments. Voice-led decisions ranged from 5% to 40% across group sizes, demonstrating why deployment composition should be reported together with aggregate coverage. The review-priority monitor also separated named decisions with 81.8% and 24.8% exact agreement ($p < 0.001$), turning decision metadata into an effective aid for human verification. Overall, the deterministic cascade provides broad coverage, traceable decisions, real-time execution, and a reproducible evaluation path for multimodal meeting intelligence.

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